

A Quick Look at New Age Trends in AI

As AI evolves, we must navigate the ethical and practical challenges it poses. Here's a summary of key AI trends and their significance in shaping the future of industries.



In the field of artificial intelligence, there is a common misconception that artificial intelligence (AI), machine learning (ML), deep learning (DL), genetic algorithms and neural networks are synonymous or do the same function but are merely matured by algorithms.

Actually, artificial intelligence (AI) is the traditional model devised to mimic human brain functions by computers using a technology-driven approach and mathematical and statistical models (e.g., probability). Later, machine learning (ML) was conceived to make programming models of brain functions

and improve the behaviour of AI models through various learning approaches, storing the results of machine learning models for future predictions. This was inspired by the use of the neural network, which is the core storage layer of the human brain. To improve the processing efficiency of this neural network-based solution, a distributed model was implemented for faster processing and simulation of various brain functions like object detection and analytical abilities.

Genetic algorithms improve deep learning models with self-learning ability and intuitiveness to maximise

the application of AIML models, reduce the effort in training them, and improve the knowledge base in neural networks. Generative AI, for example, uses large language models (LLM) to handle activities like content summarisation, content regeneration, and content extraction through self-learning capabilities. All these methods contribute to improving the ability of computer-based applications to emulate human-like thinking, reducing the effort in building and training the models, and cutting costs in time-to-market, with the flexibility to apply them in multi-domain solutions.